

JAVA IMPORTANT QUESTIONS

Computational Thinking, IPO Model, Problem Solving, Introduction to Java, Java Features:

- Go to Youtube channel listen Class Session-1 I have already posted
- Link: <https://youtu.be/COu5d3V07wE>
- Main Questions
- What is Java? Explain its importance.
Explain the major features of Java.
Why is Java called platform independent?
Explain the difference between Java and other programming languages.
What are the applications of Java?
Explain the basic working of Java.
Why is Java considered secure and robust?
Explain the meaning of “Write Once, Run Anywhere” (WORA).
What is the role of JVM in Java?
Explain why Java is widely used in software development.

2) Algorithms, Pseudocode, Flowcharts, Program Design, Stepwise Refinement

1. Algorithms

1. What is an algorithm? Explain its characteristics.
2. What are the advantages of using algorithms?
3. Write an algorithm to find the largest of two numbers.
4. Write an algorithm to check whether a number is even or odd.
5. Write an algorithm to find the largest of three numbers.
6. Write an algorithm to calculate the sum of first N natural numbers.

2. Pseudocode

7. What is pseudocode? Why is it used?
8. What are the advantages of pseudocode?
9. Write pseudocode to check whether a number is positive, negative, or zero.
10. Write pseudocode to find the factorial of a number.
11. Write pseudocode to find the sum of digits of a number.
12. Differentiate between an algorithm and pseudocode.

3. Flowcharts

13. What is a flowchart? Explain its purpose.
14. Explain the commonly used flowchart symbols.
15. Draw a flowchart to check whether a number is even or odd.
16. Draw a flowchart to find the largest of three numbers.
17. Draw a flowchart to calculate the factorial of a number.
18. What are the advantages and limitations of flowcharts?

4. Program Design

19. What is program design?
20. Explain the basic steps involved in designing a program.
21. Why is program design important before writing code?
22. Explain the relationship between problem analysis, algorithm, flowchart, and program implementation.

5. Stepwise Refinement

23. What is stepwise refinement?
24. Explain the concept of **top-down design**.
25. What are the advantages of stepwise refinement?
26. Explain stepwise refinement with a suitable example.
27. How does stepwise refinement help in solving complex problems?

Java Program Structure

Explained in Youtube channel Go to through this link: <https://youtu.be/COu5d3V07wE>

Java Architecture, JVM, JDK, JRE, Java Program Execution

Topics: Java Architecture, JVM, JDK, JRE & Java Program Execution

Java Architecture

1. What is Java Architecture? Explain its major components.
2. Explain the architecture of Java with a neat diagram.
3. Why is Java called platform independent?
4. Explain the role of JVM in Java Architecture.
5. What is “Write Once, Run Anywhere” (WORA)? Explain.

JVM – Java Virtual Machine

6. What is JVM? Explain its purpose.
7. Explain the major components of JVM.
8. What is the role of Class Loader in JVM?
9. What is Bytecode? Why is it important?
10. Explain the JVM memory areas.
11. What is the role of the Execution Engine?
12. How does JVM make Java platform independent?

JDK – Java Development Kit

13. What is JDK?
14. What are the major components of JDK?

15. What is the purpose of the javac compiler?

16. Differentiate between JDK and JVM.

JRE – Java Runtime Environment

17. What is JRE?

18. What are the components of JRE?

19. What is the purpose of JRE?

20. Differentiate between JDK, JRE, and JVM.

Java Program Execution

21. Explain the complete process of Java program execution.

22. What happens when we execute a .java file?

23. Explain the steps from Source Code → Compilation → Bytecode → Execution.

24. What is the difference between compilation and execution in Java?

25. Explain the role of javac and java commands.

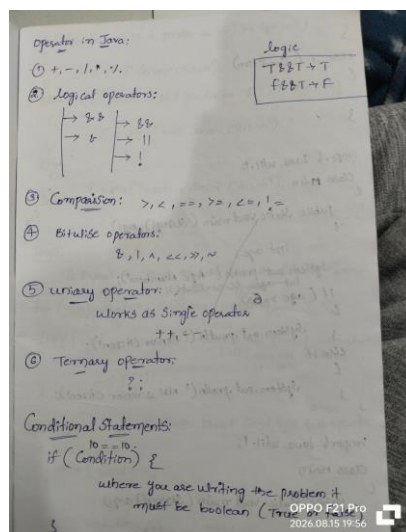
26. Why does Java use bytecode instead of directly generating machine code?

Variables, Identifiers, Literals, Data Types

Already discussed in Youtube channel go through this two links:

- 1) https://youtu.be/_pWHsxowpIY?si=-HdWPc6J3RyVaqnm
- 2) https://youtu.be/lxJub7O2_Uc?si=7b5CkAjoYO5lll0X

Operators, Expressions, Arithmetic, Relational, Logical Operators



See I have shared photo it is Operators Go through it carefully very important topic read about that article

Scanner Class, Console Input, User Input, Keyboard Interaction

Go through youtube Channel

If,if-else,else-if

Topic 1: if

1. Write a Java program to check whether a number is positive.
2. Write a Java program to check whether a number is negative.
3. Write a Java program to check whether a number is even.
4. Write a Java program to check whether a person is eligible to vote.

Topic 2: if-else

1. Write a Java program to check whether a number is even or odd.
2. Write a Java program to check whether a number is positive, negative, or zero.
3. Write a Java program to find the largest of two numbers.
4. Write a Java program to check whether a student has passed or failed.
5. Write a Java program to check whether a year is a leap year.

Topic 3: Nested if

1. Write a Java program to find the largest among three numbers.
2. Write a Java program to find the smallest among three numbers.
3. Write a Java program to calculate the grade of a student based on marks.
4. Write a Java program to check whether a number is positive, negative, or zero using nested if.
5. Write a Java program to check whether a person is eligible for voting and driving based on age.

Most Important Programs

1. Largest of 3 numbers using nested if
2. Leap year
3. Student grade calculation
4. Even or odd

5. Positive, negative, or zero
6. Largest of 2 numbers
7. Pass or fail

Switch case:

switch-case

Basic switch-case

1. Write a Java program to perform **addition, subtraction, multiplication, and division** using switch-case.
2. Write a Java program to print the **day of the week** using switch-case.
3. Write a Java program to print the **month name** based on the month number using switch-case.
4. Write a Java program to create a **simple calculator** using switch-case.

Real-World Programs

5. Write a Java program to display a **menu-driven food ordering system** using switch-case.
6. Write a Java program to display the **number of days in a month** using switch-case.
7. Write a Java program to convert **numbers 1–5 into words** using switch-case.
8. Write a Java program to implement a **menu-driven banking system** using switch-case.

Important Practice Programs

9. Write a Java program to check whether a given **character is a vowel or consonant** using switch-case.
10. Write a Java program to perform **area calculations** for Circle, Rectangle, and Square using switch-case.
11. Write a Java program to perform **temperature conversion** using switch-case.
12. Write a Java program to implement a **menu-driven arithmetic operation** using switch-case.

Must Practice:

- Simple Calculator
- Day of Week
- Month Name
- Number of Days in Month

- Vowel/Consonant
- Menu-driven Calculator
- Area Calculation using switch-case

Introduction to Loops & Pattern Printing

1: Introduction to Loops

1. Write a Java program to print numbers from **1 to N** using a loop.
2. Write a Java program to print numbers from **N to 1**.
3. Write a Java program to print all **even numbers from 1 to N**.
4. Write a Java program to print all **odd numbers from 1 to N**.
5. Write a Java program to find the **sum of first N natural numbers**.
6. Write a Java program to find the **sum of even numbers from 1 to N**.
7. Write a Java program to find the **factorial of a number**.
8. Write a Java program to print the **multiplication table** of a given number.
9. Write a Java program to count the **number of digits** in a number.
10. Write a Java program to find the **sum of digits** of a number.
11. Write a Java program to reverse a given number.
12. Write a Java program to check whether a number is a **palindrome**.
13. Write a Java program to check whether a number is an **Armstrong number**.
14. Write a Java program to check whether a number is **prime**.
15. Write a Java program to print all **prime numbers from 1 to N**.
16. Write a Java program to generate the **Fibonacci series** up to N terms.

Pattern Printing:

Right triangle

Inverted triangle

Number patterns

Pyramid

Hollow square

Diamond pattern